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Ho Chi Minh City University of Technology and Engineering
HCM-UTE

TUTORIAL

# Passive and Active Current Mirrors

*Basic, Cascode, and Active Current Mirrors, and Common-Mode Characteristics of the Differential Pair*

Department of Electronics and Information Engineering | Ho Chi Minh City University of Technology and Engineering (HCM-UTE), Vietnam

**Based on Design of Analog CMOS Integrated Circuits, Chapter 5 (Razavi)**

OUTLINE

# Roadmap for This Lecture

*Four sections, from the basic mirror to common-mode behavior*

## I Basic Current Mirrors

- MOSFET in saturation as a current source; pitfalls of resistor biasing
- The basic diode-connected mirror and its W/L ratio

## II Cascode Current Mirrors

- Channel-length modulation error in the basic mirror
- Self-biased cascode construction and the headroom/accuracy trade-off

## III Active Current Mirrors

- Fixed current-source load vs. true active mirror load
- Large-signal transfer characteristic and small-signal Gm, Rout, gain

## IV Common-Mode Characteristics

- Common-mode gain via the tail impedance RSS
- CMRR formula and the benefit of a cascode tail source

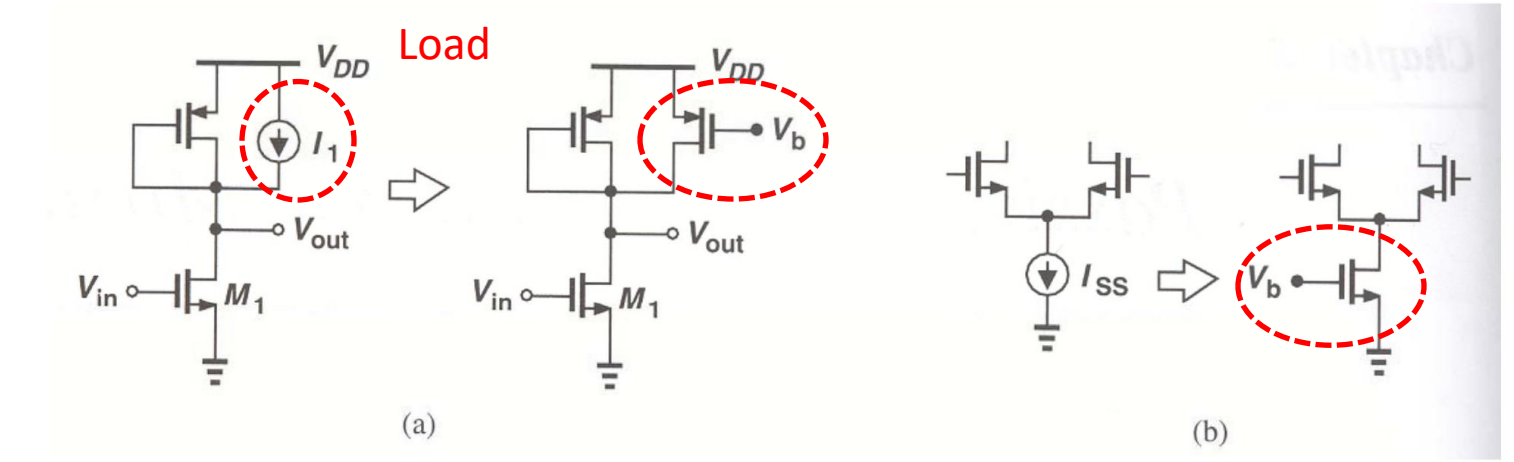
I

# Basic Current Mirrors

- MOSFET in saturation as a current source; pitfalls of resistor biasing
- The basic diode-connected mirror and its W/L ratio
- Cascaded and multi-ratio mirrors for biasing amplifier stages

# A Saturated MOSFET as a Current Source

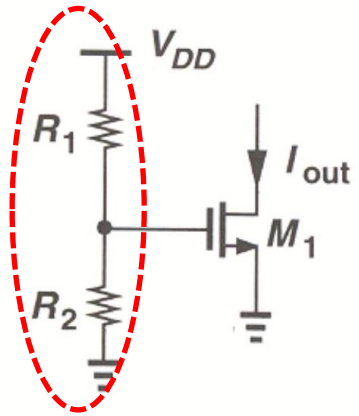
- A common-source stage's ideal load I1 (or a differential pair's tail current ISS) is, in practice, built from a MOSFET biased in saturation, using a diode-connected device to set the gate bias Vb
- This motivates the current-mirror technique developed through the rest of the chapter to generate accurate, matched bias voltages for such current sources




MOSFET in saturation region can be a current source

# Resistive-Divider Biasing of a Current Source (§5.1)

- Gate voltage for M1 set by a resistor divider R1, R2 from VDD: I_OUT = (μnCox/2)(W/L)[R2/(R1+R2)·VDD − VTH]²
- This crude scheme makes I_OUT depend directly on VDD, on VTH (temperature/process), and on the absolute accuracy of on-chip resistors — all poorly controlled
- Motivates the current-mirror approach, which relies on device matching rather than absolute component values

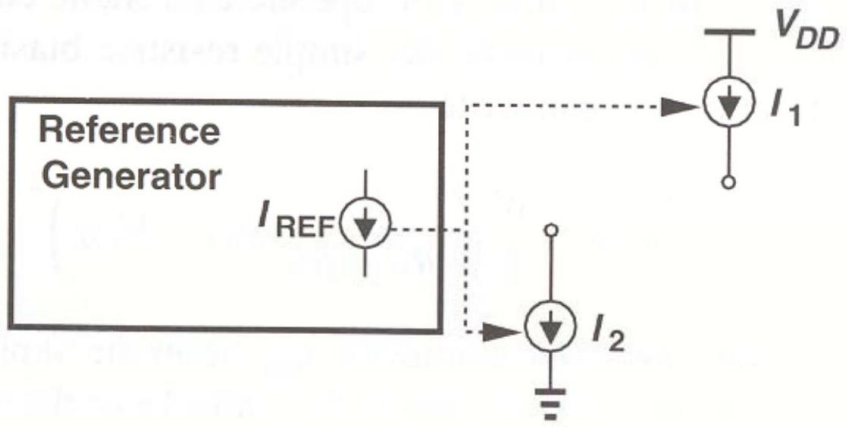


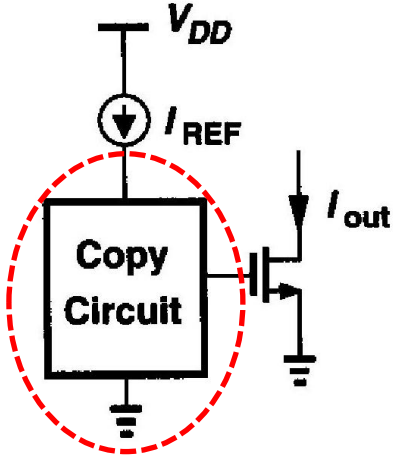

$$I_{OUT} \approx \frac{u_n C_{ox}}{2} \frac{W}{L} \left( \frac{R_2}{R_2 + R_1} V_{DD} - V_{TH} \right)^2$$

# The Current-Mirror Concept

- **A reference generator** produces a stable I_REF, which a 'copy circuit' (diode-connected transistor) mirrors into one or more matched transistors to bias I1, I2, ... elsewhere in the chip
- Because the reference and mirror transistors share VTH and μnCox (same process, matched layout), I_OUT/I_REF depends only on their (W/L) size ratio
- This ratio-based design is far less sensitive to VDD or absolute process parameters than the resistor-divider approach

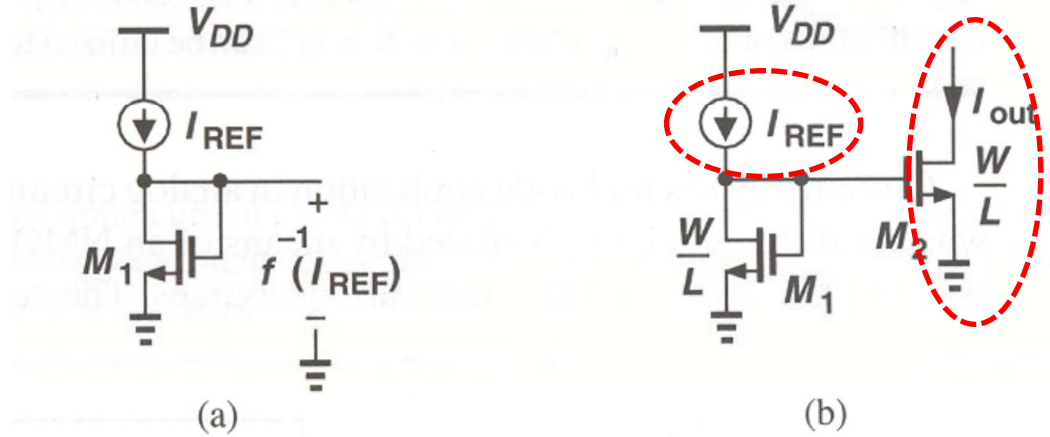
➢**Use of a reference to generate various currents**
➢**Current mirrors copy the currents.**



# Basic Current Mirror — Quantitative Analysis

- M1 is diode-connected (gate tied to drain), carrying I_REF and setting a common VGS on the shared gate line
- I_REF = $(\mu nCox/2)(W/L)_1(VGS-VTH)^2$, I_OUT = $(\mu nCox/2)(W/L)_2(VGS-VTH)^2$ — same VGS on both devices, so $I_{OUT}/I_{REF} = (W/L)_2/(W/L)_1$
- Neglecting channel-length modulation, the output/reference current ratio is set purely by the geometric sizing of M1 and M2



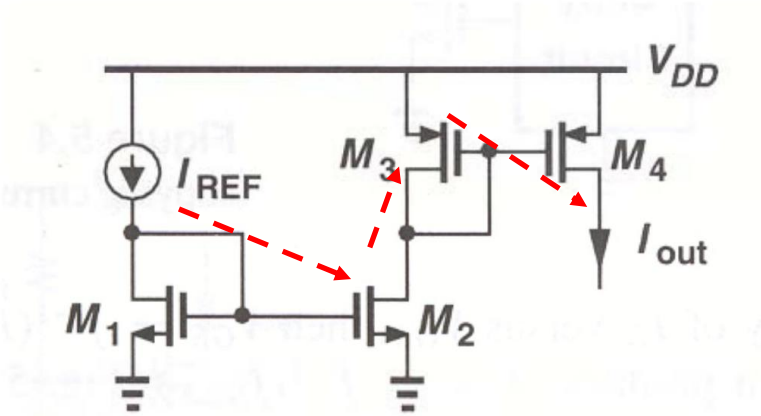

$$I_{REF} = \frac{u_n C_{ox}}{2}\left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2, I_{OUT} = \frac{u_n C_{ox}}{2}\left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH})^2$$

➢**Neglecting CLM, $I_{OUT}$ is identical to $I_{REF}$.**

# Ex. 5.1 — Cascaded NMOS/PMOS Mirror Stages (*)

- I_REF sets I_D2 through an NMOS mirror (M1 diode-connected, M2 mirror); I_D2 then flows into a PMOS mirror (M3 diode-connected, M4 mirror), producing I_D4 = I_out
- I_D2 is mirrored from I_REF, and I_D4 is mirrored from I_D3 (= I_D2)
- A single I_REF can be relayed through cascaded mirrors of different type (NMOS then PMOS) to bias any node in the circuit, near VDD or near ground

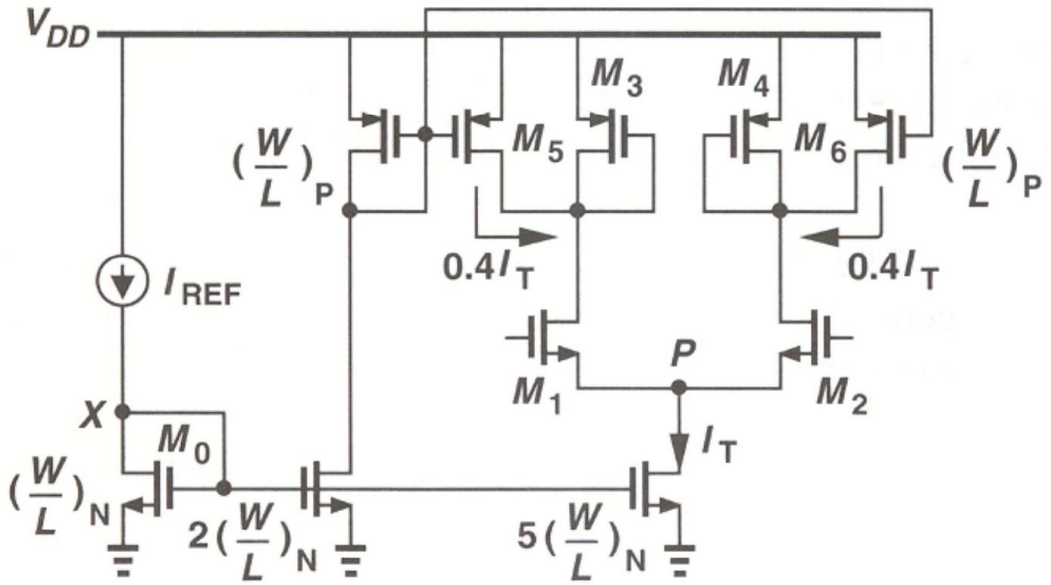
**Example 5.1) find the drain current of M4**




ID2 is mirrored from IREF and ID4 is mirrored from ID3

# Multi-Ratio Current Mirrors Biasing a Differential Amplifier

- A single I_REF (through diode-connected M0, size (W/L)_N at node X) is scaled by ratioed NMOS branches — 2(W/L)_N and 5(W/L)_N — to generate larger tail-type currents
- PMOS mirrors ((W/L)_P) further copy I_REF into 0.4·I_T branches that bias M5, M6, which set the differential pair's tail current I_T
- Because every branch current is a fixed ratio of the same I_REF, the entire bias network tracks together under temperature and process variation

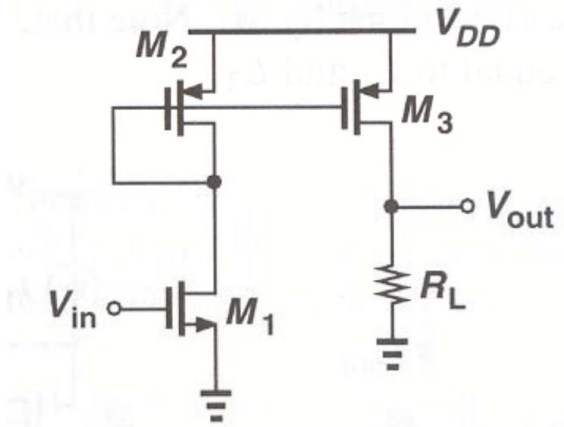



Current mirrors used to bias a differential amp

# Ex. 5.2 — Gain of a CS Stage with Mirror + Resistive Load

- M1 (common-source stage) drives diode-connected M2, whose gate mirrors M1's signal current into M3 with ratio $(W/L)_3/(W/L)_2$; M3's drain current then flows through RL to form Vout
- $A_V = g_{m1} \cdot R_L \cdot [(W/L)_3/(W/L)_2]$ — the mirror ratio acts as a current gain applied to the CS stage's signal current before it reaches RL
- A precursor to the active current-mirror load of §5.3, where RL is replaced by the mirror transistor's own ro (typically with a 1:1 ratio)

**Example 5.2) calculate the small-signal gain**




$$A_V = g_{m1} \cdot R_L \cdot \left[(W/L)_3/(W/L)_2\right]$$

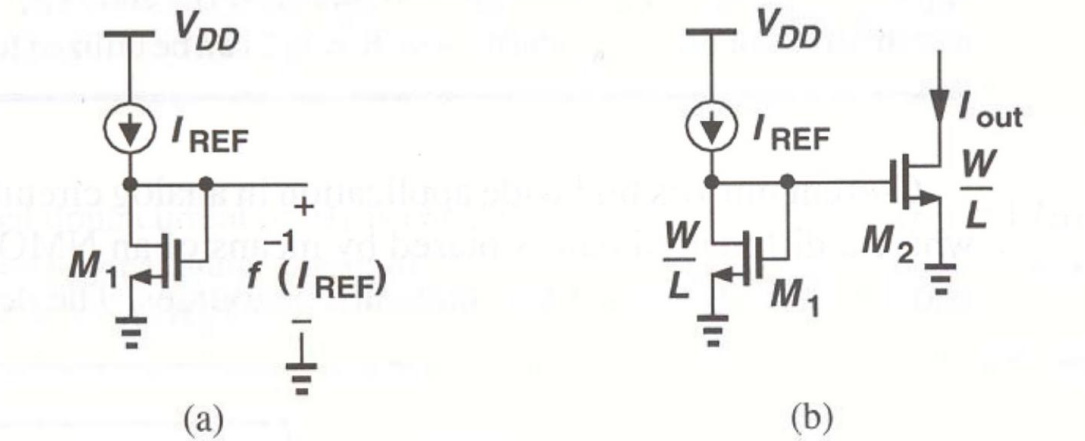
II

# Cascode Current Mirrors

- Channel-length modulation error in the basic mirror
- Self-biased cascode construction and the headroom/accuracy trade-off
- Low-voltage cascode techniques for reduced headroom

# Channel-Length Modulation in the Basic Mirror (§5.2)

- Including channel-length modulation: I_REF = (μnCox/2)(W/L)₁(VGS−VTH)²(1+λVDS1), I_OUT = (μnCox/2)(W/L)₂(VGS−VTH)²(1+λVDS2)
- M1 is diode-connected so VDS1=VGS is fixed, but VDS2 depends on the rest of the circuit; generally VDS2≠VDS1, so I_OUT deviates from the ideal ratio
- Motivates the cascode current mirror, which forces VDS2 closer to VDS1 while also raising the output resistance



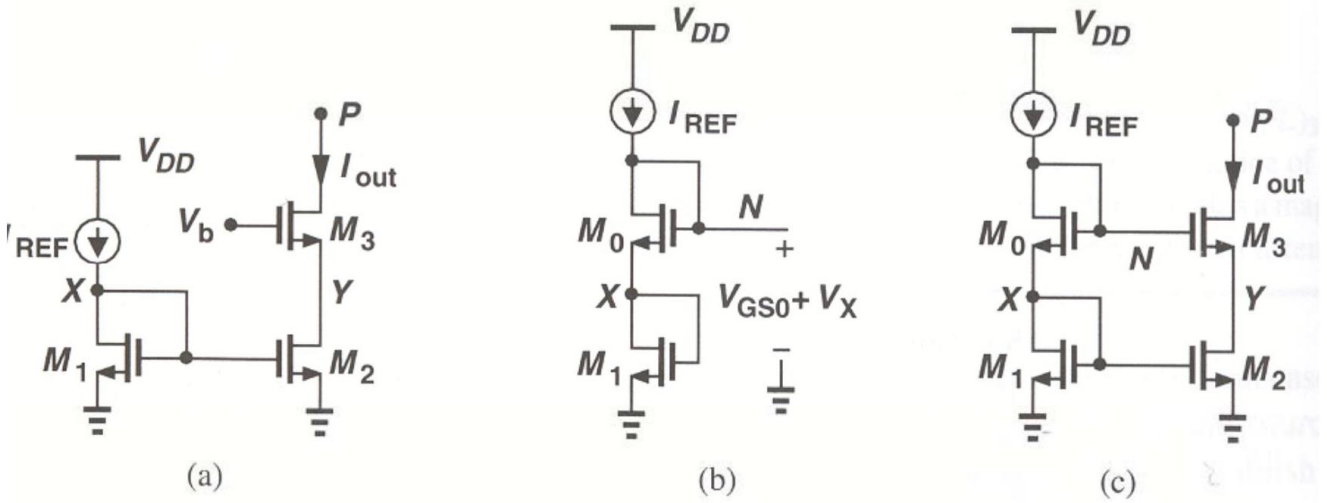

$$I_{REF} = \frac{u_n C_{ox}}{2}\left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2 (1 + \lambda \cdot V_{DS1})$$

$$I_{OUT} = \frac{u_n C_{ox}}{2}\left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH})^2 (1 + \lambda \cdot V_{DS2})$$

➢**$I_{OUT}$ is different from $I_{REF}$ due to the channel length modulation**

# Cascode Current Mirror — Three-Step Construction

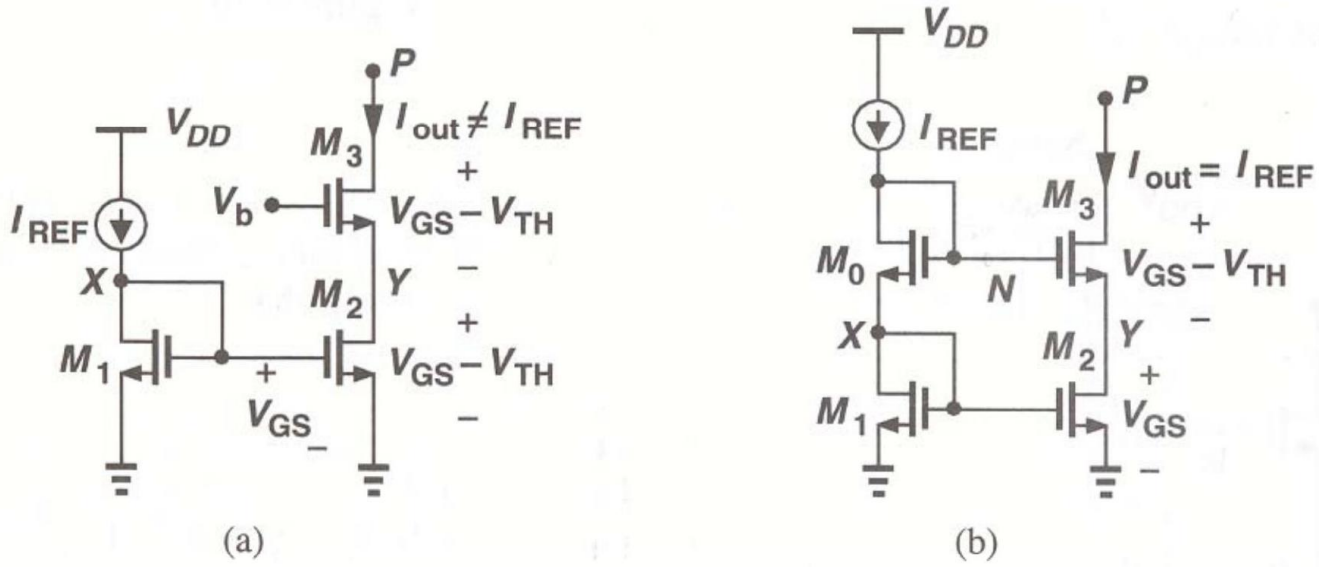
- (a) Cascode current source: M1 (diode-connected, sets I_REF) plus an output branch M2–M3 stacked, with M3's gate at a fixed external Vb, giving very high output resistance
- (b) Stacking a diode-connected M0 above M1 self-generates the cascode bias at node N = VGS0+VX, removing the need for an external Vb
- (c) Combining both: node N directly biases the output-branch cascode M3, forming a fully self-biased cascode current mirror




**(a) Cascode current source**
**(b) Modification of mirror circuit to generate the cascode bias voltage**
**(c) Cascode current mirror**

# Headroom vs. Accuracy Trade-off in Cascode Mirrors

- (a) M3 biased at the minimum external Vb for saturation (VDS3=VGS3−VTH) minimizes headroom and maximizes output swing, but I_out ≠ I_REF since Vb does not track I_REF
- (b) The self-biased stack (M0–M1) gives each device a full VGS−VTH (more headroom), but I_out = I_REF because the mirror branch's VDS closely tracks the reference branch's
- This is the classic accuracy-vs-swing trade-off in cascode current-mirror design

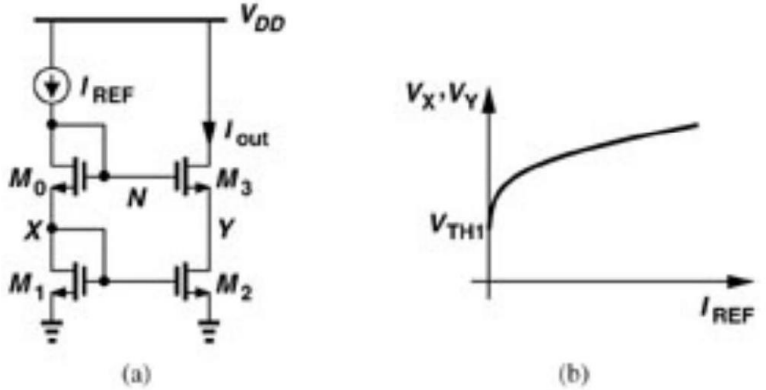



**(a) Cascode current source with minimum headroom voltage**
**(b) The headroom is consumed by a cascode mirror circuit but $I_{OUT}$ is more identical to $I_{REF}$**

# Ex. 5.3 — Maximum I_REF in a Self-Biased Cascode Mirror

- V_N = VGS0+VGS1 = √(2I_REF/μCox)·[√(L/W)₀+√(L/W)₁] + VTH0+VTH1, with headroom for the I_REF source requiring VDD−V_N ≥ 0.5 V
- Solving gives I_REF,max = (μCox/2)(VDD−0.5−VTH0−VTH1)² / [√(L/W)₀+√(L/W)₁]²
- Larger W/L (smaller √(L/W)) permits a higher I_REF for the same VDD — a direct trade-off between current, device area, and supply voltage

Ex5.3) sketch Vx and Vy as a function of Iref
If Iref requires 0.5V to operate as a current source, what is its maximum value?



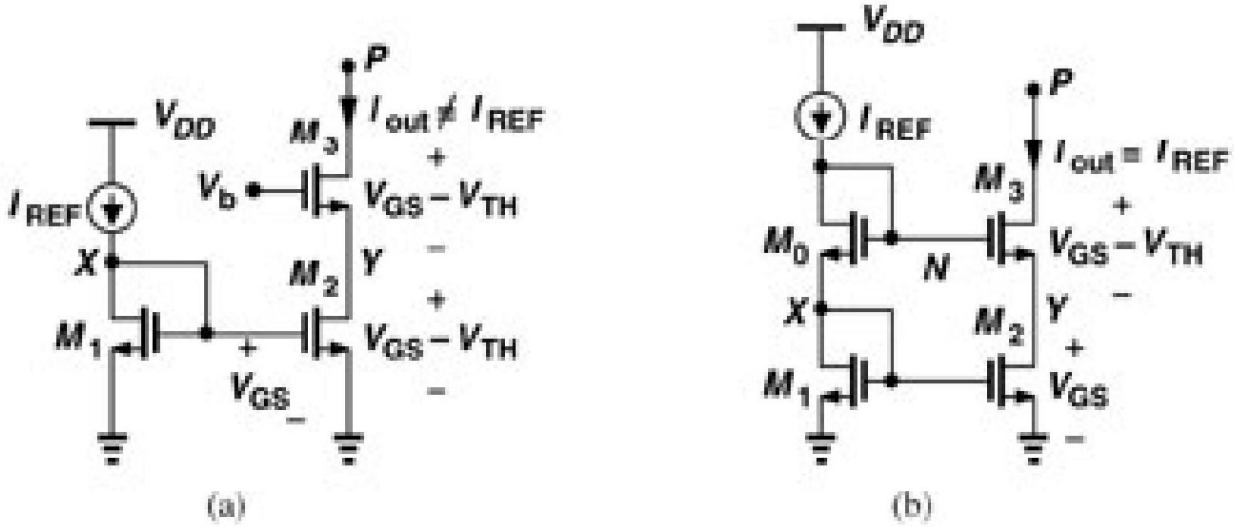

$$V_N = V_{GS0} + V_{GS1} = \sqrt{\frac{2I_{ref}}{uCox}}\left[\sqrt{\left(\frac{L}{W}\right)_0} + \sqrt{\left(\frac{L}{W}\right)_1}\right] + V_{T0} + V_{T1}$$

$$V_{DD} - V_N = 0.5$$

$$I_{ref,\max} = \frac{uCox}{2}\frac{(V_{DD} - 0.5 - V_{T0} - V_{T1})^2}{\left[\sqrt{\left(\frac{L}{W}\right)_0} + \sqrt{\left(\frac{L}{W}\right)_1}\right]^2}$$

# Recap — Two Cascode Bias Configurations

- (a) Cascode current source with minimum headroom voltage: external Vb minimizes headroom but gives I_out ≠ I_REF
- (b) Headroom consumed by a cascode mirror: self-biased M0–M1 stack gives I_out ≈ I_REF at the cost of extra headroom
- This recap of the accuracy-vs-headroom trade-off sets up Ex. 5.4's detailed I–V characteristic at the mirror's output node

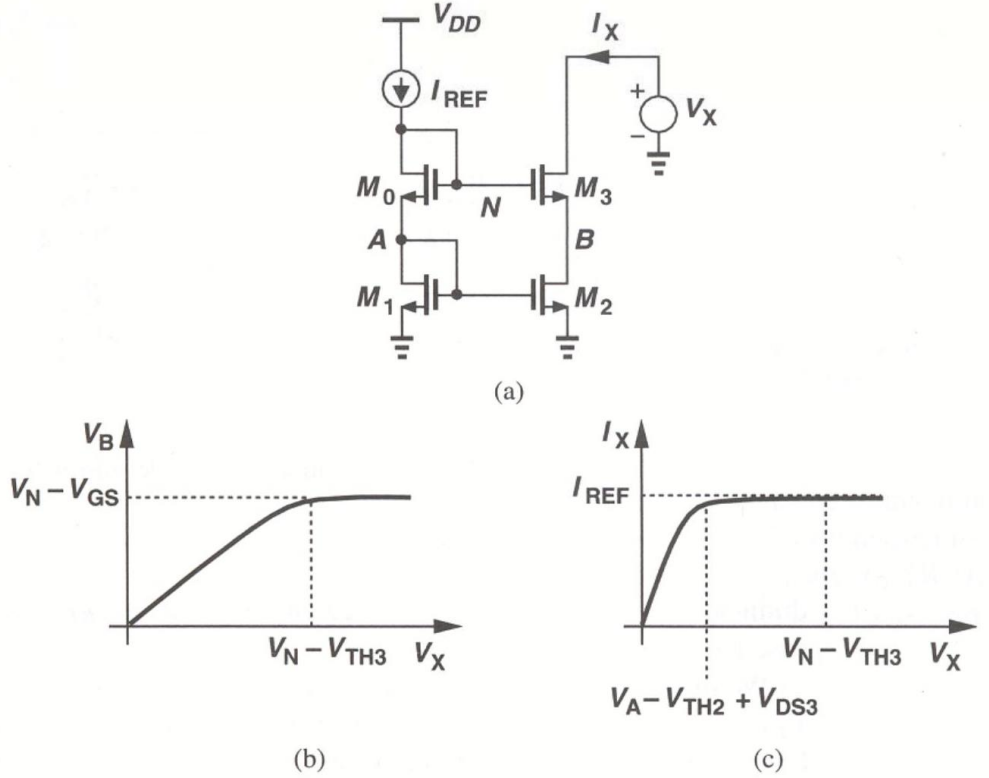



(a) Cascode current source with minimum headroom voltage
(b) Headroom consumed by a cascode mirror

# Ex. 5.4 — Output I–V Characteristic of a Cascode Mirror

- Sweeping a test voltage VX at the output and measuring IX and node VB traces the mirror's usable output-compliance range
- VB rises linearly with VX while M3 is in triode, then saturates at V_N−VGS once M3 enters saturation (VX > V_N−VTH3)
- IX rises quickly and saturates at I_REF once VX exceeds roughly V_A−VTH2+VDS3 — defining the minimum output voltage for accurate current-source behavior

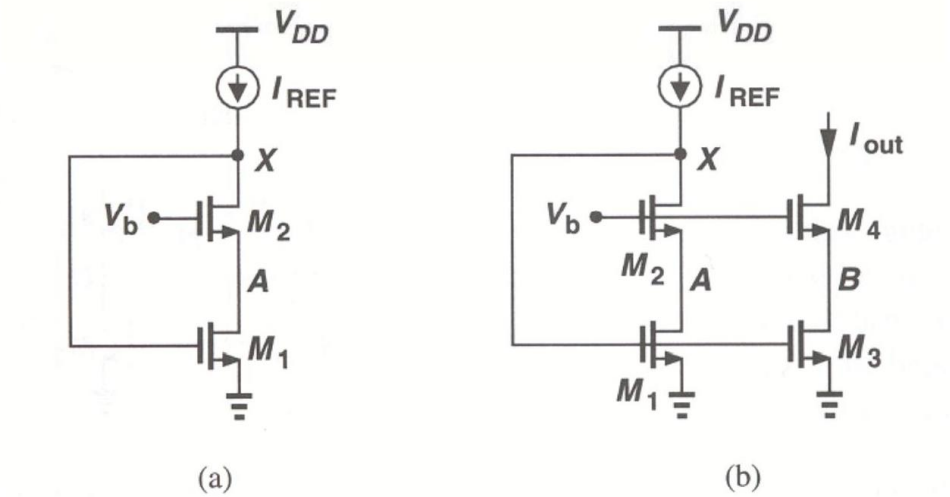
**Example 5.4)**
**sketch Ix and VB as VX varies**



# Low-Voltage Cascode Current Mirror — Bias Condition

- For M1 and M2 to both stay in saturation: VGS1 > Vb−VTH (M2 saturation) and Vb−VGS2 > VGS1−VTH (M1 saturation), giving the valid window VGS1+VTH > Vb > VGS2+VGS1−VTH
- This window exists only if VGS2 < 2VTH — a condition that can fail at low supply voltages or high bias currents where VGS2 is large
- Branch (b) extends the idea to a complete mirror (M3, M4 sharing the same Vb) at reduced headroom versus the standard cascode

➢**Low-voltage Cascode Current Mirror**

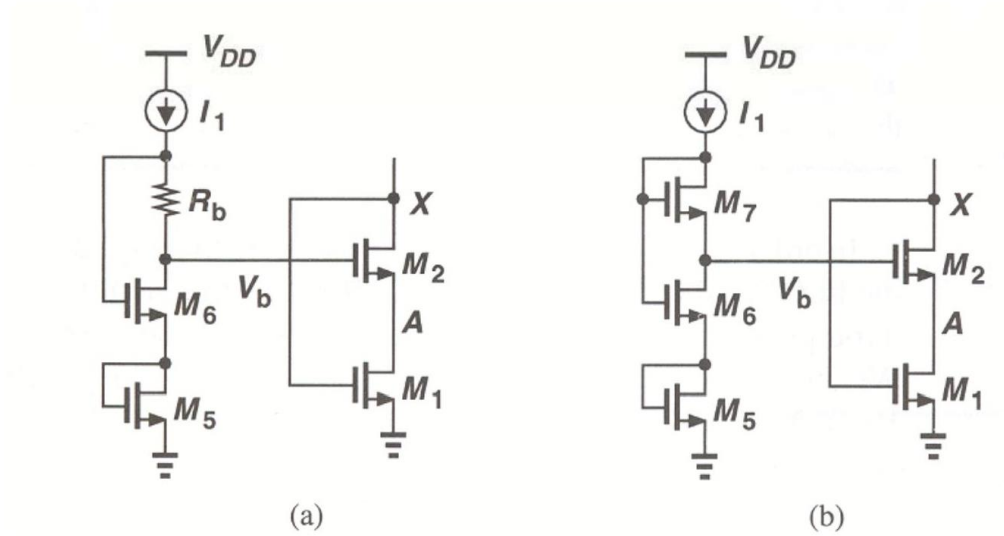



➢**M2 in saturation; VGS1>Vb-VTH**
➢**M1 in saturation; Vb-VGS2>VGS1-VTH**
➢**VGS1+VTH>Vb>VGS2+VGS1-VTH**
➢**VGS1+VTH>VGS2+VGS1-VTH => VTH>VGS2-VTH**

# Generating the Low-Voltage Cascode Bias Voltage Vb

- Both circuits stack M6 (top) and M5 (bottom, grounded) carrying I1 to produce VGS5+VGS6, then reduce it to a usable Vb
- (a) A resistor Rb inserted in the stack drops I1·Rb: Vb = VGS5+VGS6−I1·Rb, subject to Vb > VGS2+VGS1−VTH
- (b) A very large M7 replaces Rb, giving VGS7≈VTH7, so Vb=VGS5+VGS6−VTH7 — avoids resistor mismatch and tracks VTH with process/temperature

➢**Low-voltage Cascode Current Mirror**



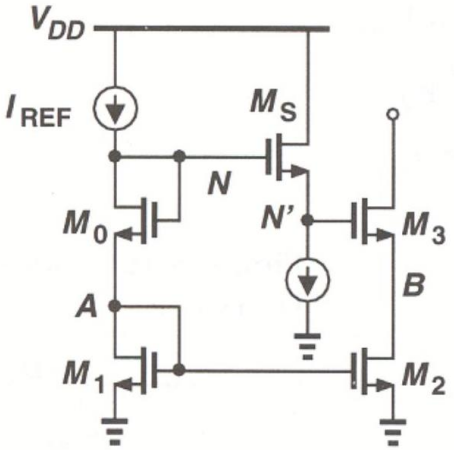

**Generation of gate voltage Vb**
**(a) Set Vb=VGS5+VGS6-I1*RB (> VGS2+VGS1-VTH)**
**(b) Set Vb=VGS5+VGS6-VTH7 (>VGS2+VGS1-VTH)**

**$(W/L)_7$ is large**

# Refined Low-Voltage Cascode with Auxiliary Bias (Ms)

- An auxiliary transistor Ms and a small current source at node N' fine-tune the cascode gate voltage of the output-branch transistor M3
- With VGS0 = VGS3 by design, VB = VGS1+VGS0−VTH3−VGS3 = VGS1−VTH3, placing M2 right at the edge of triode — near-minimum possible headroom
- Achieves near-minimum headroom while retaining the high output resistance of the cascode structure, even though VDS1 ≠ VDS2

➢**Low-voltage Cascode Current Mirror**




➢**if VGS0=VGS3**
➢**VB=VGS1+VGS0-VTH3-VGS3=VGS1-VTH3**
**(VGS at $M_S$ is biased as VTH3.)**
➢**M2 is at the edge of triode region.**
➢**VDS1 differs from VDS2 but Rout becomes very large**



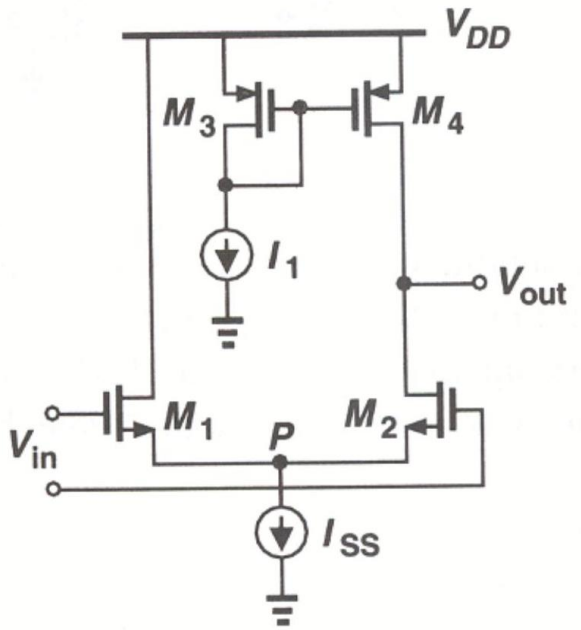

# Active Current Mirrors

- Fixed current-source load vs. true active mirror load
- Large-signal transfer characteristic and small-signal Gm, Rout, gain
- The 5-transistor OTA vs. a simple CS stage

# §5.3 Active Current Mirrors — Differential Pair with Current-Source Load

- M1, M2 form a differential pair with tail current ISS at node P; PMOS load M3, M4 share a common gate biased by a separate current source I1 — not yet a current mirror
- Vout is taken single-ended at M2's drain
- This fixed-load configuration serves as the baseline for comparison before the load is replaced by a true current mirror (from slide 23 onward)

**Differential pair with current-source load**

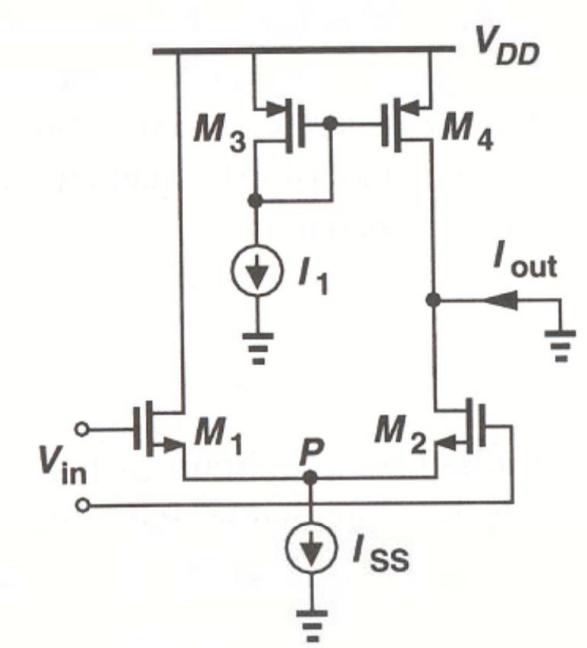



(a)

# Effective Transconductance with a Fixed Current-Source Load

- Gm ≡ Iout/Vin (output shorted to small-signal ground) = gm/2 — only half of a single transistor's gm
- Because M3, M4 are fixed current sources with no feedback, M1's branch signal current change does not reach the output — only M2's own drain-current change contributes to Iout
- This factor-of-2 loss motivates the active current-mirror load, which recovers the full gm (shown on slide 27)

**Differential pair with current-source load**

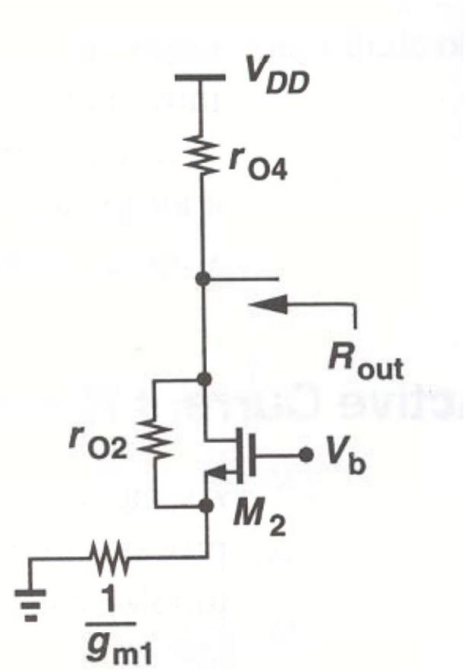



**Circuit for calculating Gm** **=> Gm=gm/2**

# Setting Up Rout for the Current-Source Load

- Looking into Vout: rO4 (M4's output resistance) appears upward to VDD (small-signal ground); rO2 appears downward, in parallel with M2 itself
- M2's source (tail node P) is represented by an equivalent resistance ≈ 1/gm1, since Vin=0 makes M1 look like a grounded-gate device from node P
- Structurally identical to a source-degenerated CS stage with RS=1/gm1, whose general result is recalled on the next slide

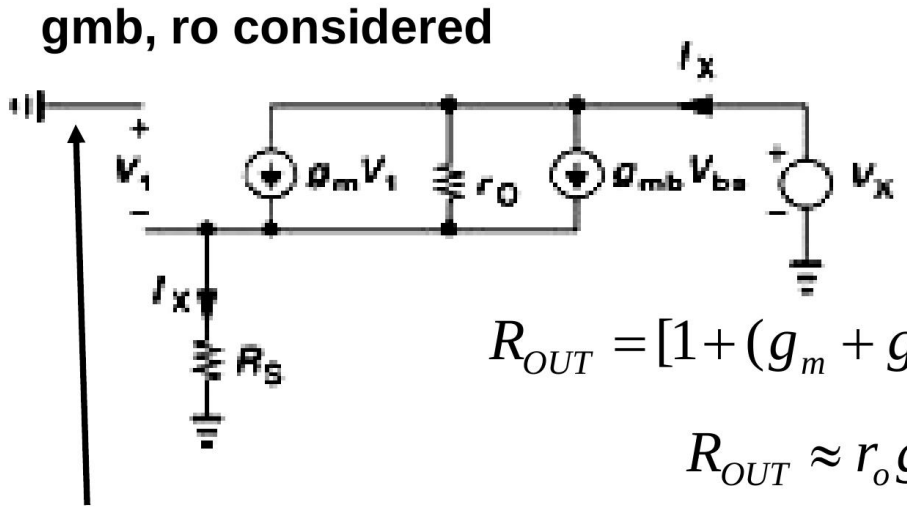
**Differential pair with current-source load**




**Circuit for calculating Rout**

# Review (Ch. 3) — CS Output Resistance with Source Degeneration

- Full result including body effect and finite ro: R_OUT = [1+(gm+gmb)ro]·RS + ro
- When (gm+gmb)ro is large, R_OUT ≈ ro·gm·RS; for the special case RS = 1/gm, R_OUT ≈ 2ro
- General degenerated-CS gain formula: Av = −Gm(RD∥R_OUT) — reused throughout the rest of the chapter




$$R_{OUT} = [1 + (g_m + g_{mb}) r_o] R_S + r_o$$

$$R_{OUT} \approx r_o g_m \cdot R_S$$

$$if\ R_S = 1/g_m \Rightarrow R_{OUT} \approx r_o g_m \cdot (1/g_m) + r_o$$

$$A_v = -G_m (R_D \parallel R_{OUT})$$

# Gain of the Differential Pair with Fixed Current-Source Load

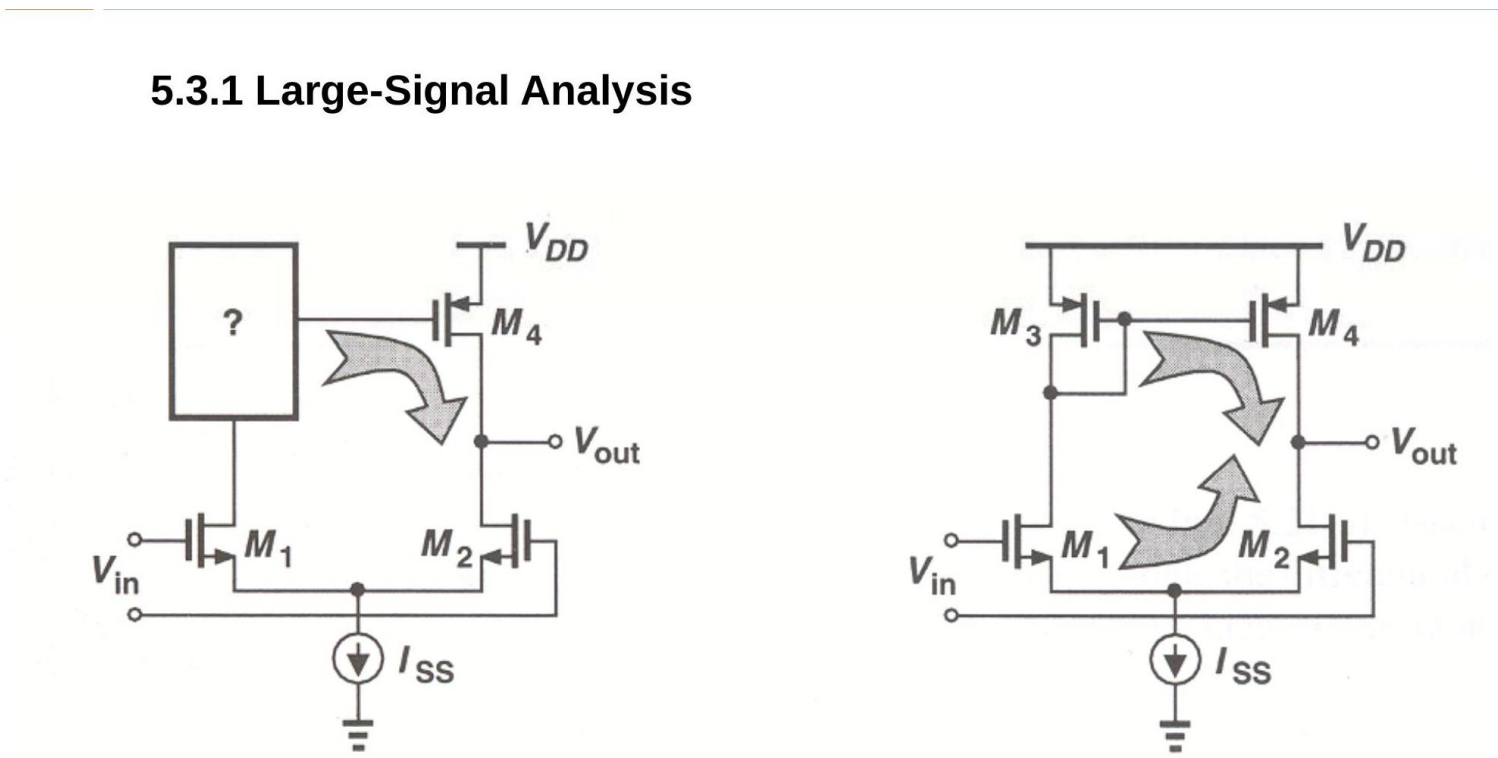
- Combining Gm = gm1/2 (slide 19) with Rout = 2rO2∥rO4 (from the RS=1/gm result of slide 21) gives A_V ≈ (gm1/2)·[2rO2∥rO4]
- Roughly half the gain of the active current-mirror-load configuration derived later (A_V ≈ gm1,2·(rO2∥rO4), slide 29) — direct quantitative motivation for the mirror load

**Differential pair with current-source load**

$$A_V \approx \frac{gm_1}{2}\left[2r_{o2} \parallel r_{o4}\right]$$

# §5.3.1 Large-Signal Analysis — Replacing the Load with a True Mirror

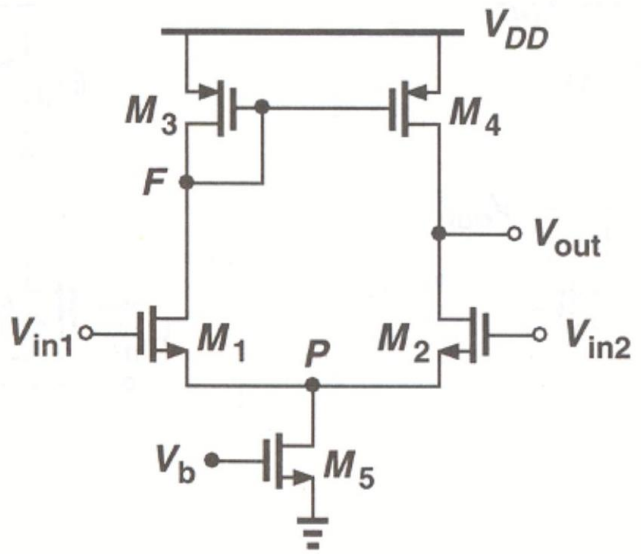
- The undefined load block '?' (fixed current sources M3–M4 of slides 18–22) is replaced by an actual current mirror: M3 diode-connected, M4 the mirror transistor
- M1's signal current is copied through M3→M4 and adds directly to M2's signal current at Vout, since M4 sources current into the output node while M2 sinks it — the two contributions add rather than cancel
- This mechanism restores the full transconductance (Gm=gm) lost in the fixed-load configuration

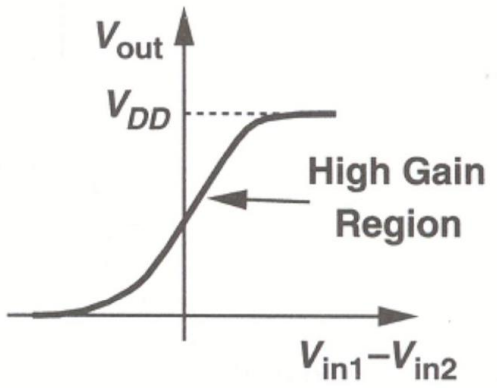


# Complete Circuit and DC Transfer Characteristic

- M3 (diode-connected, node F) and M4 form the PMOS mirror load; M1, M2 (with independent inputs Vin1, Vin2) form the input pair, with tail source M5 setting ISS at node P
- The Vout vs. (Vin1−Vin2) curve is the classic sigmoid: Vout saturates near VDD or drops low as the tail current is fully steered to one side
- The steepest region near (Vin1−Vin2)=0 is the 'High Gain Region' — the intended small-signal operating point, whose slope equals A_v derived later

**5.3.1 Large-Signal Analysis**

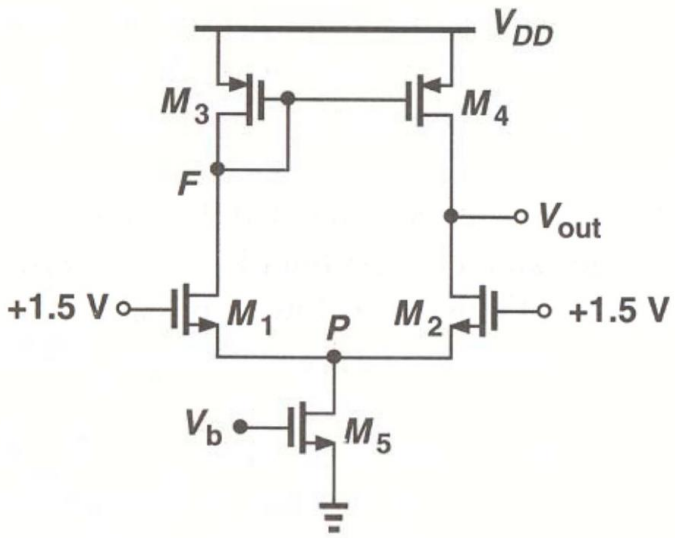


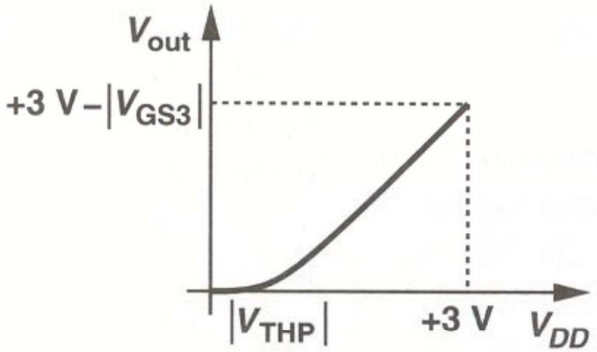
# Ex. 5.5 — Vout vs. VDD Sweep (Power-Up Behavior)

- With Vin1=Vin2=+1.5 V fixed (differential input = 0; a power-up/common-mode scenario), VDD is swept from 3 V down to 0 V
- Vout ≈ 0 while VDD < |VTHP| (PMOS load cannot conduct); above |VTHP|, Vout rises roughly linearly, reaching VDD−|VGS3| at VDD = 3 V
- The mirror load's internal feedback makes Vout track VDD rather than stay independent of it — a power-supply-rejection concern

**Example 5.5) sketch the output voltage as VDD varies from 3V to 0**




(a)

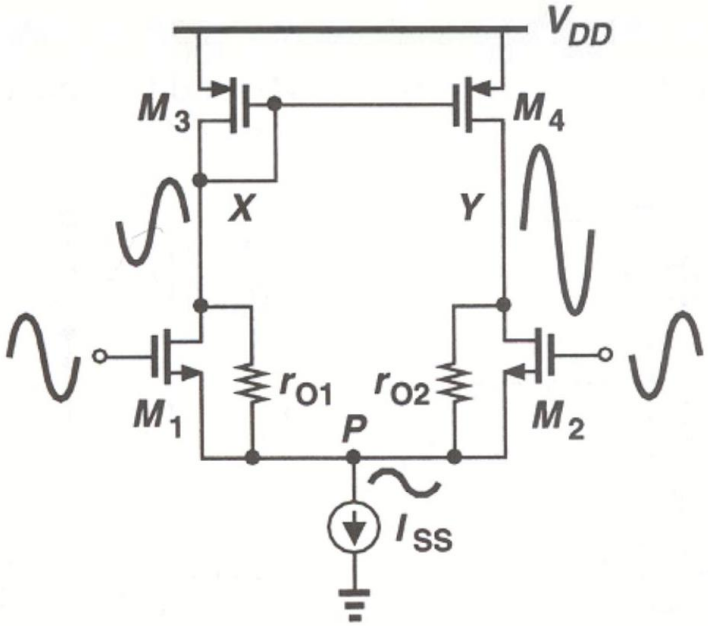



(b)

# §5.3.2 Small-Signal Analysis — Setup

- Small-signal excitation is applied at M1, M2's gates; nodes X, Y (drains of M1, M2) carry rO1, rO2 respectively
- For a purely differential input, tail node P behaves as a virtual ground, so the tail source's rO5 can be neglected in this analysis
- Governing framework: |A_v| ≈ Gm·Rout, with Gm derived next (slide 27) and Rout derived after (slide 28)

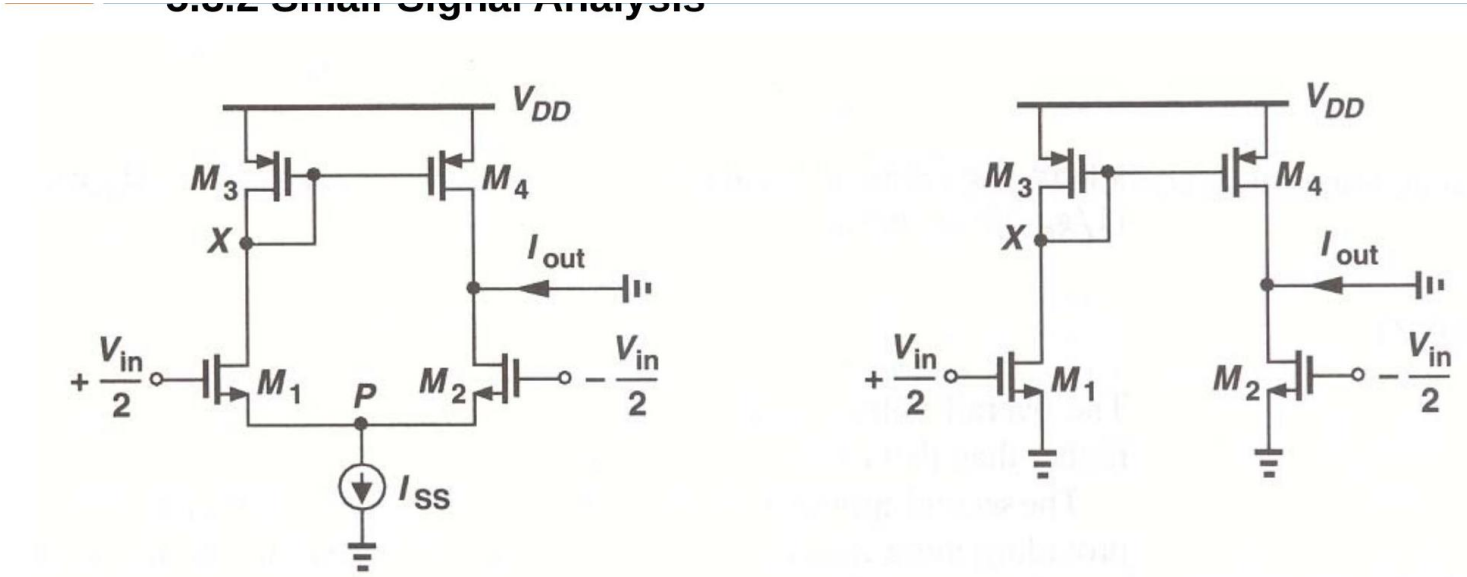
**5.3.2 Small-Signal Analysis**




$$|A_v| \approx G_m R_{out}$$

# Deriving Gm for the Active Current-Mirror Load

- With ±Vin/2 applied differentially: ΔI_D1=ΔI_D3=ΔI_D4=gm1,2·(Vin/2) (M3 diode-connected passes M1's current, mirrored 1:1 to M4), while ΔI_D2=−gm1,2·(Vin/2)
- At Vout: I_out = ΔI_D2−ΔI_D4 = −gm1,2·Vin, so Gm = −gm1,2 — the full transconductance, not halved
- Confirms the benefit over the fixed current-source load (slide 19, Gm=gm/2): the mirror relays M1's signal current to add constructively with M2's at the output

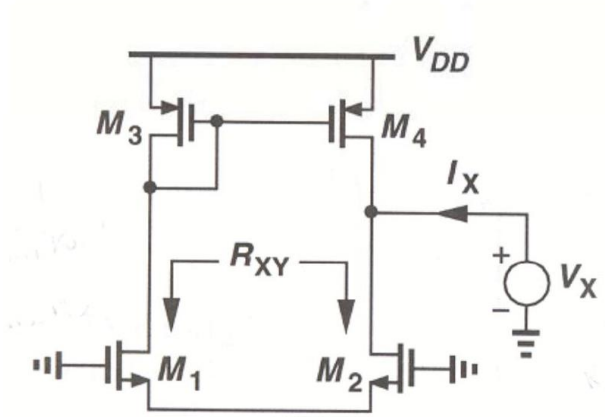


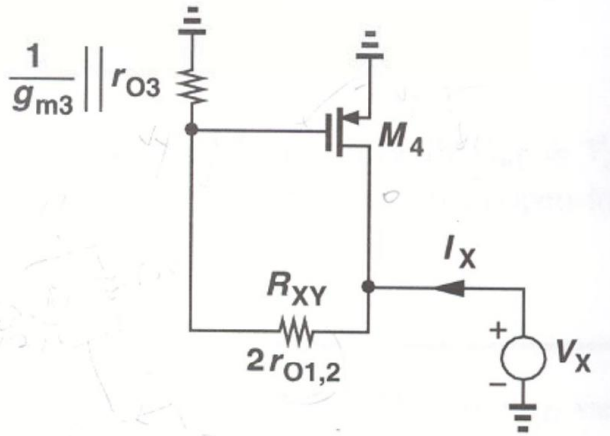

$$\Delta I_{D1} = \Delta I_{D3} = \Delta I_{D4} = g_{m1,2} V_{in} / 2, \quad \Delta I_{D2} = -g_{m1,2} V_{in} / 2$$

$$I_{out} = \Delta I_{D2} - \Delta I_{D4} = -g_{m1,2} V_{in}, \Rightarrow G_m = -g_{m1,2}$$

# Deriving Rout for the Active Current-Mirror Load

- A test source VX at Vout drives current through two paths: directly through rO4, and through M2 into the shared X-Y node (R_XY≈2rO1,2), part of which is mirrored back via (1/gm3)∥rO3 into M4
- I_X = 2V_X/(2rO1,2+1/gm3) + V_X/rO4; since 2rO1,2 ≫ (1/gm3)∥rO3, the first term is negligible
- Result: R_out ≈ rO2∥rO4 — the mirror's feedback path through M3-M4 contributes negligibly to the output resistance




➢**Rout Calculation**

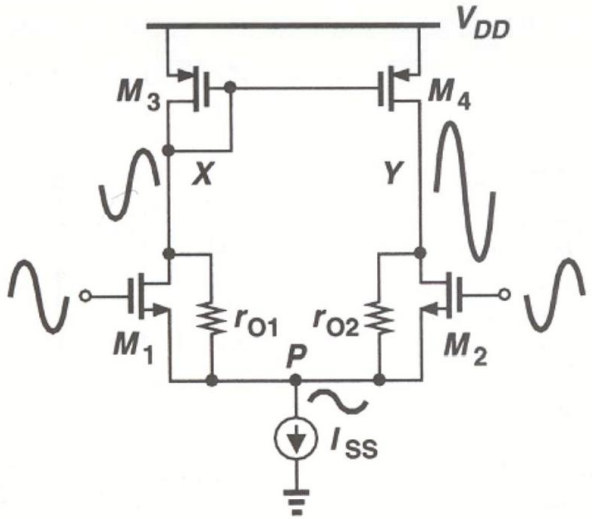
$$I_X = 2\frac{V_X}{2r_{o1,2} + 1/g_{m3}} + \frac{V_X}{ro4}$$

**current through 1/gm3 mirrored by M4**

$$R_{out} \approx r_{o2} \| r_{o4}\,, \quad (2r_{o1,2} >> [1/g_{m3}] \| r_{o3})$$

# Small-Signal Gain — Active Current-Mirror Load

- Combining Gm = gm1,2 (slide 27) with Rout = rO2∥rO4 (slide 28): A_V ≈ gm1,2·(rO2∥rO4)
- About twice the gain of the fixed current-source load (slide 22), since the full Gm dominates despite a slightly smaller Rout
- This differential pair + active mirror load + tail source forms the ubiquitous '5-transistor OTA,' a basic building block of CMOS op-amps

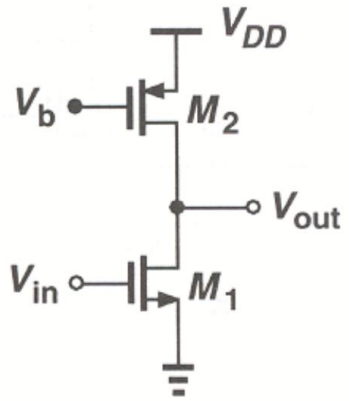



$$A_v \approx g_{m1,2}(r_{o2} \parallel r_{o4})$$

# Ex. 5.6 — Simple CS Stage vs. Differential Pair Gain

- A single CS stage (M1) with a fixed current-source load (M2, gate at fixed Vb, not diode-connected) has A_v ≈ gm1,2(rO2∥rO4) — algebraically identical in form to the differential pair with active-mirror load (slide 29)
- Slide note: 'Voltage gain is the same as the differential amplifier. Noise is not suppressed in this circuit.'
- The real advantage of the differential pair is not raw gain but common-mode/noise rejection — motivating the following section

**Example 5.6 calculate the small signal voltage**




$$A_v \approx g_{m1,2}(r_{o2} \parallel r_{o4})$$

**Voltage gain is the same with the differential amp.**
**Noise not be suppressed in this circuit**

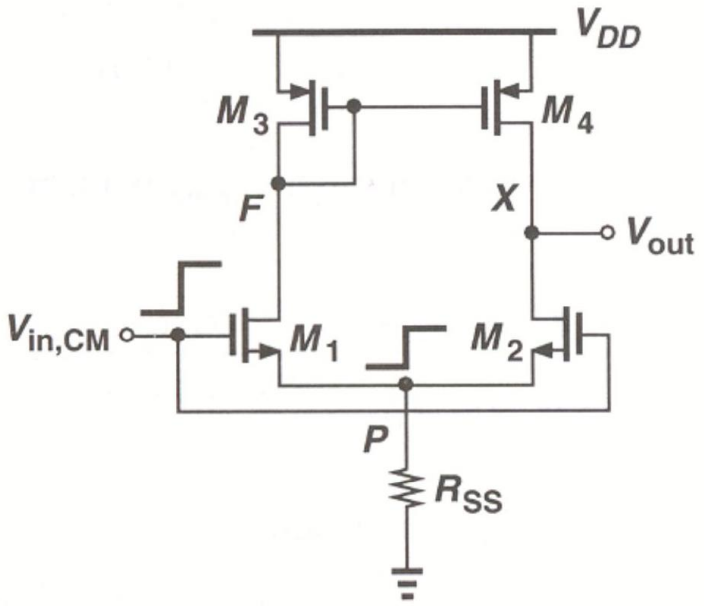
# IV

# Common-Mode Characteristics

- Common-mode gain via the tail impedance RSS
- CMRR formula and the benefit of a cascode tail source
- CMRR degradation from gm mismatch between input devices

# Common-Mode Characteristics — Setup

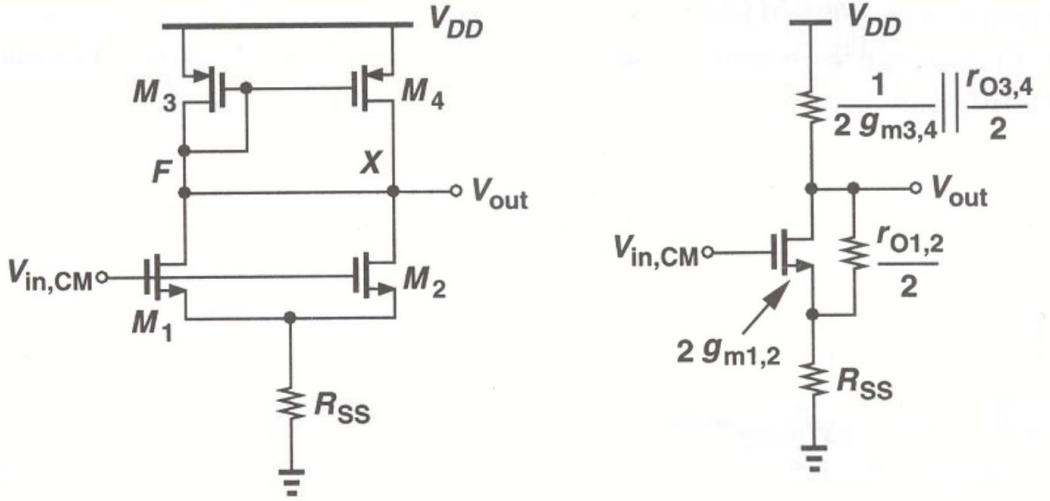
- Both M1, M2 gates now receive the same common-mode signal Vin,CM, and the ideal tail source ISS is replaced by a finite tail resistance RSS
- Common-mode gain is defined A_CM = ΔVout/ΔVin,CM
- With RSS→∞ (ideal tail source), a common-mode input change does not alter the tail current, so A_CM=0; finite RSS makes A_CM nonzero




$$A_{CM} = \frac{\Delta V_{out}}{\Delta V_{in,CM}}$$

# Deriving A_CM via the Common-Mode Half-Circuit

- By symmetry, the half-circuit has effective transconductance 2gm1,2, output resistance rO1,2/2, unchanged tail resistance RSS, and mirror-equivalent load (1/(2gm3,4))∥(rO3,4/2)
- A_CM ≈ [−gm1,2/(1+2gm1,2RSS)]·(1/gm3,4) — structurally analogous to the source-degenerated gm/(1+gmRS) formula from Chapter 3
- Larger gm1,2·RSS (better tail current source) drives A_CM toward zero, confirming the intuition from the previous slide




$$A_{CM} \approx \frac{-2g_{m1,2}}{1+2g_{m1,2}R_{SS}} \cdot \frac{1}{2g_{m3,4}} = \frac{-g_{m1,2}}{1+2g_{m1,2}R_{SS}} \cdot \frac{1}{g_{m3,4}}$$

# Deriving CMRR

- CMRR ≡ |A_DM/A_CM|; substituting A_DM=gm1,2(rO1,2∥rO3,4) (slide 29) and A_CM from the previous slide, the gm1,2 terms cancel
- CMRR = gm3,4·(rO1,2∥rO3,4)·(1+2gm1,2RSS) — proportional to gm3,4, the parallel output resistance, and critically to the tail resistance RSS
- This is why practical designs almost always use a cascode tail current source (§5.2 of this chapter) — its much higher RSS directly boosts CMRR

$$CMRR = \left|\frac{A_{DM}}{A_{CM}}\right|$$

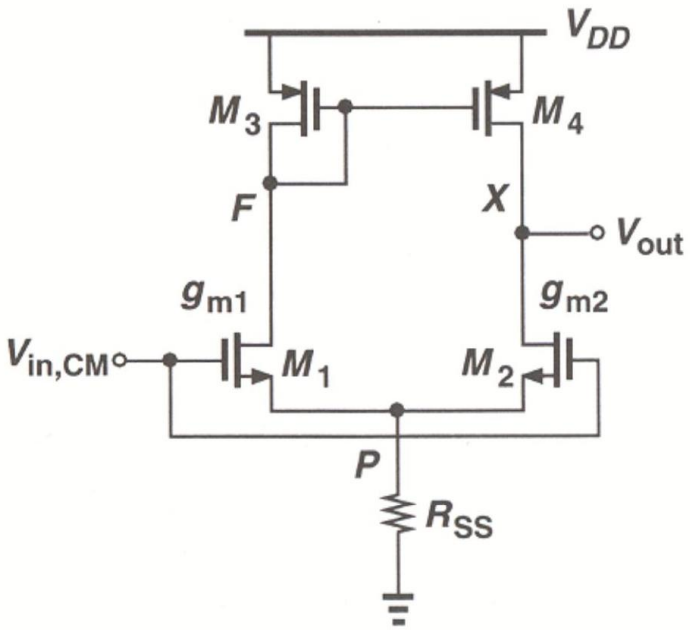
$$= g_{m1,2}(r_{o1,2} \parallel r_{o3,4})\frac{g_{m3,4}(1+2g_{m1,2}R_{SS})}{g_{m1,2}}$$

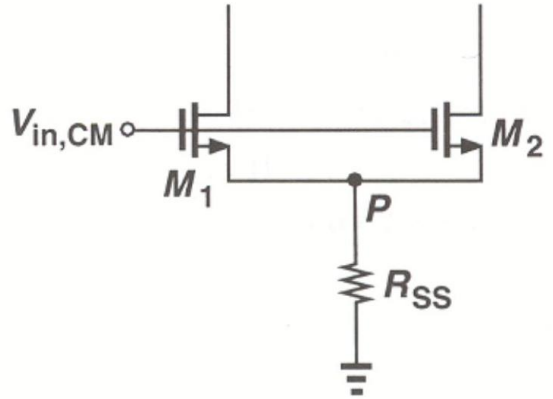
$$= g_{m3,4}(r_{o1,2} \parallel r_{o3,4})(1+2g_{m1,2}R_{SS})$$

# gm Mismatch — Setup

- Considers a second non-ideality: M1, M2 with mismatched transconductances gm1≠gm2 (random or systematic W/L or VTH mismatch)
- The reduced half-circuit keeps only Vin,CM, M1, M2, and RSS — preparing for the quantitative derivation on the next slide
- Even with an ideal cascode tail source (high RSS), gm mismatch independently degrades CMRR — motivating common-centroid layout

➢gm mismatch



# Quantitative Effect of gm1≠gm2 on Common-Mode Output

- Tail voltage under common-mode drive: ΔV_P = ΔVin,CM·RSS/[RSS+1/(gm1+gm2)]; branch currents scale as gm1/(gm1+gm2) and gm2/(gm1+gm2)
- If gm1=gm2, the mirror cancels the two branch currents exactly at Vout; if gm1≠gm2, the mirrored ΔI_D4 no longer cancels ΔI_D2
- ΔVout=[ΔI_D4−ΔI_D2]·rO4 ≠ 0 even for a purely common-mode input — the mechanism behind common-mode-to-differential-mode conversion

$$\Delta V_P = \Delta V_{in,CM}\frac{g_{m1}+g_{m2}}{1+(g_{m1}+g_{m2})R_{SS}}R_{SS} = \Delta V_{in,CM}\frac{R_{SS}}{R_{SS}+\dfrac{1}{g_{m1}+g_{m2}}}$$

$$\Delta I_{D1} = g_{m1}(\Delta V_{in,CM}-\Delta V_P) = \frac{\Delta V_{in,CM}}{R_{SS}+\dfrac{1}{g_{m1}+g_{m2}}}\frac{g_{m1}}{g_{m1}+g_{m2}}$$

$$\Delta I_{D2} = g_{m2}(\Delta V_{in,CM}-\Delta V_P) = \frac{\Delta V_{in,CM}}{R_{SS}+\dfrac{1}{g_{m1}+g_{m2}}}\frac{g_{m2}}{g_{m1}+g_{m2}}$$

$$\Delta I_{D4} = g_{m4}\left[(1/g_{m3})\,\|\,r_{o3}\right]\Delta I_{D1}$$

$$\Delta V_{out} = \left[\Delta I_{D4}-\Delta I_{D2}\right]r_{o4}$$